

Algebra 1
Unit 9
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

Alejandro's work when solving the equation $y = ax^2 - 6x + 9$ for a is shown. Use Alejandro's work to answer questions 1 and 2.

$$y = ax^2 - 6x + 9$$

$$y - (-6x + 9) = ax^2 - 6x + 9 - (-6x + 9)$$

$$y + 6x + 9 = ax^2$$

$$\frac{y + 6x + 9}{x^2} = \frac{ax^2}{x^2}$$

$$\frac{y + 6x + 9}{x^2} = a$$

1. Circle the step where Alejandro made an error.
2. Correct Alejandro's error and the steps that follow to show how to correctly isolate a in the equation. Use the space provided above to show the correct steps.

$$y + 6x - 9 = ax^2$$

$$\frac{y+6x-9}{x^2} = \frac{ax^2}{x^2} \rightarrow a = \frac{y+6x-9}{x^2}$$

Solve each equation for the designated variable. Show your work.

3. Solve $-3(x + 5)(x + m) = y$ for m .

$$\frac{-3(x + 5)(x + m)}{-3(x + 5)} = \frac{y}{-3(x + 5)}$$

$$x + m = \frac{y}{-3(x + 5)}$$

$$m = \frac{y}{-3(x + 5)} - x$$

4. Solve $y = 2x^2 + bx - 9$ for b .

$$y + 9 = 2x^2 + bx - 9 + 9$$

$$y + 9 - 2x^2 = 2x^2 + bx - 2x^2$$

$$\frac{y - 2x^2 + 9}{x} = \frac{bx}{x}$$

$$\frac{y - 2x^2 + 9}{x} = b$$

5. Solve $y = \frac{8}{3}(x + 2)^2 + r$ for r. $y - \frac{8}{3}(x + 2)^2 = \frac{8}{3}(x + 2)^2 + r - \frac{8}{3}(x + 2)^2$ $y - \frac{8}{3}(x + 2)^2 = r$	6. Solve $y = -3(x - 2)^2 + 4$ for x. $y - 4 = -3(x - 2)^2 + 4 - 4$ $\frac{y - 4}{-3} = \frac{-3(x - 2)^2}{-3}$ $\sqrt{\frac{y - 4}{-3}} = \sqrt{(x - 2)^2}$ $\pm \sqrt{\frac{y - 4}{-3}} = x - 2 \rightarrow x = 2 \pm \sqrt{\frac{y - 4}{-3}}$
7. Solve $y = ax^2 + 7x - 9$ for a. $y - (7x - 9) = ax^2 + 7x - 9 - (7x - 9)$ $\frac{y - 7x + 9}{x^2} = \frac{ax^2}{x^2}$ $\frac{y - 7x + 9}{x^2} = a$	8. Solve $y = k(x - 3)^2 - 7$ for k. $y + 7 = k(x - 3)^2 - 7 + 7$ $\frac{y + 7}{(x - 3)^2} = \frac{k(x - 3)^2}{(x - 3)^2}$ $\frac{y + 7}{(x - 3)^2} = k$
9. Solve $y = x^2 - 12x + 4w$ for w. $y - (x^2 - 12x) = x^2 - 12x + 4w - (x^2 - 12x)$ $\frac{y - x^2 + 12x}{4} = \frac{4w}{4}$ $\frac{y - x^2 + 12x}{4} = w$	10. Solve $y = a(5x - 1)(x + 1)$ for a. $\frac{y}{(5x - 1)(x + 1)} = \frac{a(5x - 1)(x + 1)}{(5x - 1)(x + 1)}$ $\frac{y}{(5x - 1)(x + 1)} = a$